$$\hat{\sigma}_z(\tau) = \mathrm{e}^{\mathrm{i}\hat{H}_S\tau}\hat{\sigma}_z\mathrm{e}^{-\mathrm{i}\hat{H}_S\tau} = \hat{\sigma}_z\cos\left(\Delta_0\tau\right) - \hat{\sigma}_y\sin\left(\Delta_0\tau\right). \tag{5.141}$$